$$(D) = \sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'} \gamma_{im}\gamma_{li} = \gamma_{im}\gamma_{li}\frac{t(t+1)(t+2)}{6}.$$

Plugging these expressions back into eq. (A.24), we get:

$$\begin{aligned}
-\nabla_{W_k}\mathbb{E}_{\boldsymbol{x}\sim\mathcal{P}}\left[L|\mathcal{P}\right] = \frac{c\pi_l\left(\mathcal{P}_{m1}-\frac{1}{2}\right)}{t(\lambda-1)^2}\Bigg[&\frac{\beta_{1m}\beta_{l1}-\beta_{2m}\beta_{l2}}{\lambda-1}\underbrace{\left(\frac{\lambda-(t+1)\lambda^{t+1}+t\lambda^{t+2}}{(1-\lambda)^2}-\frac{1-\lambda^t}{(1-\lambda^{-1})(1-\lambda)}+t\frac{1}{\lambda-1}\right)}_{(I)}\\
&+\frac{\beta_{1m}\gamma_{l1}-\beta_{2m}\gamma_{l2}}{(\lambda-1)^2}\underbrace{\left(\lambda^2\frac{1-\lambda^t}{1-\lambda}-\lambda\frac{t(t+3)}{2}+\frac{t(t+1)}{2}\right)}_{(II)}\\
&+\frac{\beta_{l1}\gamma_{1m}-\beta_{l2}\gamma_{2m}}{\lambda-1}\underbrace{\left(t\frac{\lambda}{1-\lambda}-\frac{\lambda^2-\lambda^{t+2}}{(1-\lambda)^2}-\frac{t(t+1)}{2}\right)}_{(III)}\\
&+\left(\gamma_{1m}\gamma_{l1}-\gamma_{2m}\gamma_{l2}\right)\underbrace{\frac{t(t+1)(t+2)}{6}}_{(IV)}\Bigg].
\end{aligned} \tag{A.25}$$

Finally, by taking expectation over a, b (the randomness of the transition matrix), we get for the update of W_k with learning rate η:

$$\left(W_k^{(1)}\right)_{m,l} = \left(W_k^{(0)}\right)_{m,l} - \eta\nabla_{W_k}\mathbb{E}_{a,b}\left[\mathbb{E}_{\boldsymbol{x}\sim\mathcal{P}}\left[L|\mathcal{P}\right]\right]. \tag{A.26}$$

We consider the 4 cases separately. Case $m, l = 1, 1$:

$$\begin{aligned}
\left(W_k^{(1)}\right)_{1,1} = c + \frac{\eta c}{t}\mathbb{E}_{a,b}\Bigg[\frac{(b-1)(a-\frac{1}{2})}{(\lambda-1)^3}\Bigg[&\frac{(a-1)(a-b)}{\lambda-1}(I)+\frac{2(a-1)(b-1)}{(\lambda-1)^2}(II)\\
&+\frac{2(a-1)(b-1)}{\lambda-1}(III)+(b-1)(b-a)(IV)\Bigg]\Bigg]
\end{aligned} \tag{A.27}$$

Case $m, l = 1, 2$:

$$\begin{aligned}
\left(W_k^{(1)}\right)_{1,2} = c + \frac{\eta c}{t}\mathbb{E}_{a,b}\Bigg[\frac{(a-1)(a-\frac{1}{2})}{(\lambda-1)^3}\Bigg[&\frac{(1-b)(a-b)}{\lambda-1}(I)+\frac{2(a-1)(b-1)}{(\lambda-1)^2}(II)\\
&-\frac{2(b-1)^2}{\lambda-1}(III)+(b-1)(b-a)(IV)\Bigg]\Bigg]
\end{aligned} \tag{A.28}$$

Case $m, l = 2, 1$:

$$\begin{aligned}
\left(W_k^{(1)}\right)_{2,1} = c + \frac{\eta c}{t}\mathbb{E}_{a,b}\Bigg[\frac{(b-1)(\frac{1}{2}-b)}{(\lambda-1)^3}\Bigg[&\frac{(1-a)(a-b)}{\lambda-1}(I)-\frac{2(a-1)(b-1)}{(\lambda-1)^2}(II)\\
&+\frac{2(a-1)^2}{\lambda-1}(III)+(a-1)(b-a)(IV)\Bigg]\Bigg]
\end{aligned} \tag{A.29}$$

Case $m, l = 2, 2$:

$$\begin{aligned}
\left(W_k^{(1)}\right)_{2,2} = c + \frac{\eta c}{t}\mathbb{E}_{a,b}\Bigg[\frac{(a-1)(\frac{1}{2}-b)}{(\lambda-1)^3}\Bigg[&\frac{(b-1)(a-b)}{\lambda-1}(I)-\frac{2(a-1)(b-1)}{(\lambda-1)^2}(II)\\
&-\frac{2(a-1)(b-1)}{\lambda-1}(III)+(a-1)(b-a)(IV)\Bigg]\Bigg]
\end{aligned} \tag{A.30}$$

See Figure 6 (left) for an empirical estimation for some choice of hyperparameters. We observe that the gradient is symmetric, so it is going to be $\left(W_k^{(1)}\right)_{2,1} = \left(W_k^{(1)}\right)_{1,2}$, and the value in the diagonal is the same, i.e. $\left(W_k^{(1)}\right)_{1,1} = \left(W_k^{(1)}\right)_{2,2}$.

By only focusing on the leading order terms, (IV), we have:

$$\left(W_k^{(1)}\right)_{m,l} = c + c\eta\mathbb{E}_{a,b}\left[\frac{\pi_m\pi_l}{(\lambda-1)}(b-a)\left(\mathcal{P}_{m1}-\frac{1}{2}\right)\right]\frac{(t+1)(t+2)}{6} + O(t). \tag{A.31}$$

A calculation then shows that

$$\begin{aligned}\mathbb{E}_{a,b}\left[\frac{\pi_1\pi_1}{(\lambda-1)}(b-a)\left(\mathcal{P}_{11}-\frac{1}{2}\right)\right] &= \mathbb{E}_{a,b}\left[\frac{\pi_2\pi_2}{(\lambda-1)}(b-a)\left(\mathcal{P}_{21}-\frac{1}{2}\right)\right] \\ &= \int_0^1\int_0^1 \frac{(b-1)^2(b-a)(a-\frac{1}{2})}{(a+b-2)^3}dadb = \frac{\ln 256 - 5}{12} \approx 0.045,\end{aligned} \tag{A.32}$$

and

$$\begin{aligned}\mathbb{E}_{a,b}\left[\frac{\pi_1\pi_2}{(\lambda-1)}(b-a)\left(\mathcal{P}_{11}-\frac{1}{2}\right)\right] &= \mathbb{E}_{a,b}\left[\frac{\pi_1\pi_2}{(\lambda-1)}(b-a)\left(\mathcal{P}_{21}-\frac{1}{2}\right)\right] \\ &= \int_0^1\int_0^1 \frac{(b-1)(a-1)(b-a)(a-\frac{1}{2})}{(a+b-2)^3}dadb = \frac{7-10\ln 2}{6} \approx 0.011,\end{aligned} \tag{A.33}$$

hence the diagonal grows a constant (≈ 4) times more than the off-diagonal.

Similarly, for the expected gradient of v (with respect to $\boldsymbol{x}$ conditioned on $\mathcal{P}$), we have:

$$\begin{aligned}\mathbb{E}\left[\frac{\partial f(e)_{p,i}}{\partial v_j}\right] &= c\mathbb{E}\left[\sum_{t'=1}^{p}\mathbb{1}\left\{x_{t'}=i\right\}\sum_{s=1}^{t'}\delta_{j(t'-s+1)}\mathbb{1}\left\{j\le p\right\}\right] \\ &= c\mathbb{E}\left[\sum_{t'=1}^{p}\mathbb{1}\left\{x_{t'}=i\right\}\mathbb{1}\left\{j\le t'\right\}\mathbb{1}\left\{j\le p\right\}\right] \\ &= c\sum_{t'=j}^{p}\mathbb{P}\left[x_{t'}=i\right]\mathbb{1}\left\{j\le p\right\} \\ &= c\pi_i\left(p-j+1\right)\mathbb{1}\left\{j\le p\right\},\end{aligned} \tag{A.34}$$

and

$$
\begin{aligned}
\mathbb{E}\left[\frac{\partial f(e)_{p,x_{p+1}}}{\partial v_j}\right] &= c\sum_{t'=j}^{p}\mathbb{P}\left[x_{t'}=x_{p+1}\right]\mathbb{1}\left\{j\le p\right\} \\
&= c\sum_{t'=j}^{p}\sum_{i=1}^{k}\mathbb{P}\left[x_{t'}=x_{p+1}=i\right]\mathbb{1}\left\{j\le p\right\} \\
&= c\sum_{i=1}^{k}\sum_{t'=j}^{p}\mathbb{P}\left[x_{p+1}=i\mid x_{t'}=i\right]\mathbb{P}\left[x_{t'}=i\right]\mathbb{1}\left\{j\le p\right\} \\
&= c\sum_{i=1}^{k}\pi_i\sum_{t'=j}^{p}\left(\mathcal{P}^{p+1-t'}\right)_{ii}\mathbb{1}\left\{j\le p\right\} \\
&= c\sum_{i=1}^{k}\frac{\pi_i}{\lambda-1}\sum_{t'=j}^{p}\left(\beta_{ii}\lambda^{p+1-t'}+\gamma_{ii}\right)\mathbb{1}\left\{j\le p\right\} \\
&= c\left(\frac{b-1}{(\lambda-1)^2}\sum_{t'=j}^{p}\left((a-1)\lambda^{p+1-t'}+(b-1)\right)+\frac{a-1}{(\lambda-1)^2}\sum_{t'=j}^{p}\left((b-1)\lambda^{p+1-t'}+(a-1)\right)\right)\mathbb{1}\left\{j\le p\right\} \\
&= c\left(2\frac{(a-1)(b-1)}{(\lambda-1)^2}\frac{\lambda^{p+1-j}-1}{1-\lambda^{-1}}+\frac{(a-1)^2+(b-1)^2}{(\lambda-1)^2}(p-j+1)\right)\mathbb{1}\left\{j\le p\right\}.
\end{aligned}
\tag{A.35}
$$

Thus, the update would be:

$$
\begin{aligned}
v_j^{(1)} &= v_j^{(0)}-\eta\frac{\partial\mathcal{L}}{\partial v_j} \\
&= v_j^{(0)}+\frac{\eta}{t}\sum_{p=1}^{t}\left(\mathbb{E}_{a,b}\left[\mathbb{E}_{\boldsymbol{x}\sim\mathcal{P}}\left[\frac{\partial f(e)_{p,x_{p+1}}}{\partial v_j}\middle|\mathcal{P}\right]\right]-\frac{1}{k}\sum_{i=1}^{k}\mathbb{E}_{a,b}\left[\mathbb{E}_{\boldsymbol{x}\sim\mathcal{P}}\left[\frac{\partial f(e)_{p,x_i}}{\partial v_j}\middle|\mathcal{P}\right]\right]\right) \\
&= c+\frac{c\eta}{t}\sum_{p=j}^{t}\mathbb{E}_{a,b}\left[\frac{1}{(\lambda-1)^2}\left(\left((a-1)^2+(b-1)^2\right)(p-j+1)+2(a-1)(b-1)\frac{\lambda^{p-j+1}-1}{1-\lambda^{-1}}\right)-\frac{1}{2}(p-j+1)\right] \\
&= c+\frac{c\eta}{t}\mathbb{E}_{a,b}\left[\frac{(t-j+1)(t-j+2)}{2}\left(\frac{(a-1)^2+(b-1)^2}{(\lambda-1)^2}-\frac{1}{2}\right)+2\lambda\frac{(a-1)(b-1)}{(\lambda-1)^3}\left(\frac{\lambda-\lambda^{t-j+2}}{1-\lambda}-t+j-1\right)\right].
\end{aligned}
\tag{A.36}
$$

See Figure 6 (right) for an empirical estimation for some choice of hyperparameters. As we did in the case of the gradient of W_k, we focus on the leading order terms and we have:

$$
v_j^{(1)} = c+\frac{c\eta}{t}\frac{(t-j+1)(t-j+2)}{2}\mathbb{E}_{a,b}\left[\frac{(a-1)^2+(b-1)^2}{(\lambda-1)^2}-\frac{1}{2}\right]+O(1), \tag{A.37}
$$

and, since $\mathbb{E}_{a,b}\left[\frac{(a-1)^2+(b-1)^2}{(\lambda-1)^2}\right]=2-\ln 4>\frac{1}{2}$, we get a positive bias with quadratic dependence.

Second step. We saw that we get a bias towards a diagonal W_k after one update (A.31). Let $W_k = C'+(\rho-c')I$, where $C'=c'11^T$ and $\rho>c'$, with c' denoting the off-diagonal term and ρ the on-diagonal one. As in the first step, we can tune the margin parameter of the loss, Δ to be large enough, so